Solutions Architect I

Technical Evaluation

1. Objective

Your goal is to design, develop, and prototype a geospatial web application that demonstrates the business value of Planet's satellite imagery. As a Solutions Architect at Planet, you must bridge the gap between complex spatial data, sophisticated platforms tools and actionable client insights.

You are expected to build a functional tool that addresses a real-world use case of your choice (e.g., deforestation monitoring, urban expansion, disaster response, etc).

2. Challenge

Build and deploy (local deployment is sufficient) a web-based application or interactive dashboard to stream and analyse Planet data potentially integrating it with other spatial or non-spatial sources (e.g., Google Earth Engine, CDSE, socio-economic data, etc.).

Core Components:

- Business Logic: Implementation of logic to derive specific insights (analytics) for your use case of choice
- Map Interface: A dynamic map displaying / streaming relevant Planet imagery and analytical layers
- Analytical View: Interactive charts or visualizations (e.g., time-series analysis, NDVI histograms, change-detection trends, etc)
- Information Dashboard: A tabular or summary view providing high-level insights ● Interactivity: Components must be connected—filtering data in the table or timeline should update the map and charts accordingly

3. Technical panel presentation

During the interview, you will demo your app as if presenting on a sprint review with your team. We expect a narrative approach that combines:

- **User Story**: Define the specific problem you are solving and how Planet data and your application can help in solving it
- **Narrative**: Drive the panel through the application, connecting your technical choices back to the main business goal
- **Technical Balance:** Show that you can maintain a "high-level" perspective while being

prepared to dive into the technical details and defend your architecture.

Your solutions will be evaluated on the following criteria:

- **Functionality**: ensuring that the components and the tech stack work
- **Adherence to the assignment:** with clarity of the use case (extra point for stretch features)
- **Technical Design**: Your handling of exceptions and API constraints, data volumes, scalability and performance considerations, among others
- **Code Quality**: Structure, documentation, and your ability to defend architectural choices
- **Communication and documentation**: How effectively do you articulate the value of the solution. Your ability to bridge the gap between high-level business goals and deep technical implementation.

3. Resources & Data Access

You are encouraged to use real data workflows. To access Planet data and platform you will need to use a Planet account. For that please follow the instructions below:

1. **Create Planet Account**: Sign up for the free Planet Insights Platform Trial. 2. **Data**: Once your account has been created, we recommend focusing your analysis on one of the locations where Planet Sandbox data is available. This ensures high-quality imagery is accessible for your prototype.
3. **Planet Docs**: You can find all technical documentation regarding Planet data, platform and APIs in the documentation website.

4. Implementation Guidelines

- **Technology Stack**: You may use any programming language, although Python is preferred. For the web interface, use any tools you prefer. For rapid prototyping of maps and dashboards, frameworks like Streamlit, Dash, or Shiny are recommended.
- **AI Usage**: You are permitted to use Generative AI. Please note programming proficiency is required and you will be asked to explain the logic of your code and its structure. You must be able to clearly explain and justify every architectural decision and implementation detail.
- **Deliverables**: Please send your application as a zip file, including a README with clear instructions detailing:
    - The objective of your application
    - The core components and their functionalities
    - Data used and assumptions made
    - Clear instructions for local deployment
- **Disclaimer**: While we are not focusing on the veracity of the results, proper use of geospatial data and correct application of algorithms will impact the score. If you are aware of accuracy issues in your results, feel free to highlight them

We are excited to see how you combine technical expertise with creative problem-solving to bring Planet's data to life.